

NE555 Test 3: Manual Set and Reset

Question: How do the trigger and threshold pins control and store the output and state of the 555?

Hypothesis: If trigger pin is briefly connected to ground the output state will set and stay HIGH. The opposite will occur if threshold pin is touched to Vcc. This is because the internal SR latch stores the most recent state

Process:

1. Connect pin 8 and pin 4 to Vcc
2. Connect pin 1 to GND
3. Connect pin 3 through 220Ω resistor to grounded LED
4. Connect pin 2 to the Vcc through $10k\Omega$ pull up resistor
5. Connect a pushbutton between pin 2 and ground

6. Connect pin 6 to ground through $10k\ \Omega$ pulldown resistor
7. Connect a second pushbutton between pin 6 and V_{cc}
8. Apply power and record initial state of the LED
9. Press trigger pin button record
10. Repeat process for threshold pin
11. Repeat trigger and threshold test

Data:

Initial State :	ON
Press Trigger :	ON
Press Threshold :	OFF
Press Trigger :	ON
Press Threshold :	OFF
Press Trigger :	ON
Press Threshold :	OFF
Press Trigger :	ON
Press Threshold :	OFF

Conclusion:

- Overall this is a success
- The logic of the setup is that by using pushbuttons to provide a change in state of pin 2 and 6
 - For pin 2 it provides HIGH output when LOW so we ground the node to change state
 - For pin 6 opposite logic of pin 2 so we supply power to the node
- The biggest issue was wiring the circuit
 - Understanding the idea of an electrical node
 - Remembering the orientation of a pushbutton